TECHNICAL SPECIFICATION · v0.2.0

Digital Unconscious

Autonomous Research Pipeline

+ Continuous Learning & Human Idea Modeling

From Daily Screen Behavior → Validated, Published-Ready Research — Zero Human Intervention

Version	v0.2.0 — adds Learning Engine
Date	March 2026
Status	Concept Design
Execution Engine	Claude Code (Headless + Computer Use)
Human Touchpoints	Credential provision only
New in v0.2	Continuous Learning Engine, Human Idea Model, Meta-Prompt Evolution
License	MIT (Open Source)

"The system that learns from you becomes, over time, more you than any tool you have ever used."

1. Overview

1.1 What This Document Describes

This document specifies the Autonomous Research Pipeline — the research execution layer of the Digital Unconscious system. While the parent system passively observes screen behavior and distills creative ideas, this pipeline takes validated ideas and transforms them into rigorous, publication-ready research outputs with minimal human involvement.

Version 0.2 adds a Continuous Learning Engine that studies every research run and every human idea to make the system progressively smarter over time. The AI does not just execute — it reflects, learns, and self-improves.

> Design Principle: The pipeline is built around a single constraint —
> humans are only asked to intervene when the system genuinely cannot proceed autonomously.
> In practice this means two scenarios: (1) providing institutional credentials for paywalled resources,
> and (2) approving a final paper before submission.

1.2 Full System Architecture

```
Digital Unconscious (Parent System)
│
├── Screen Observation Layer        ← passive 24/7 monitoring
├── Compression & Summarization     ← Claude Haiku / Claude Code
├── Creative Idea Generation        ← Claude Opus, high temperature + RAG
├── Judge Agent                     ← Claude Sonnet, scoring & filtering
│                    ▲
│                    │ feedback: what kinds of ideas lead to good papers?
│                    │
└── [HIGH-SCORE IDEAS]
         │
         ▼
Autonomous Research Pipeline
         │
         ├── Stage 1 · Literature Scout
         ├── Stage 2 · Feasibility Assessment
         ├── Stage 3 · Data Acquisition (+ UI computer use)
         ├── Stage 4 · Analysis & Figure Generation
         ├── Stage 5 · Paper Drafting
         ├── Stage 6 · AI Peer Review Loop
         │
         ▼
    [COMPLETED RUN]
         │
         ▼
╔═══════════════════════════════════╗
```

```
╔  Continuous Learning Engine          ╗  ◄── NEW in v0.2
╚                                      ╝
    ┌──────────────────────────────┐
    │  Run Outcome Analyzer        │
    │  Human Idea Model            │
    │  Prompt Evolution Engine     │
    │  Domain Knowledge Expander   │
    │  Meta-Learning Scheduler     │
    └──────────────────────────────┘
            │           │
      updates       updates
            ▼           ▼
     Agent Prompts   RAG Knowledge
     & Strategies    Base
          │
          └──► feedback loops back into ALL upstream agents
```

2. Pipeline Architecture

2.1 Stages at a Glance

```
┌──────────────────────────────────────────────────────────────────────────┐
│  INPUT: Validated Idea (score > 75) from Judge Agent                     │
└──────────────────────────────────────────────────────────────────────────┘
                                   │
                                   ▼
        ┌──────────────────────────────────────────────────────────┐
        │  STAGE 1  · Literature Scout                             │
        │  Search → rank → download PDFs → extract                 │
        └──────────────────────────────────────────────────────────┘
                                   │
                                   ▼
        ┌──────────────────────────────────────────────────────────┐
        │  STAGE 2  · Feasibility Assessment                       │
        │  Evidence synthesis → go/no-go (confidence)              │
        └──────────────────────────────────────────────────────────┘
                                   │
                          GO ┌─────┴─────┐ NO-GO → archive
                             ▼
        ┌──────────────────────────────────────────────────────────┐
        │  STAGE 3  · Data Acquisition                             │
        │  Open access → credential broker → UI nav                │
        └──────────────────────────────────────────────────────────┘
                                   │
                                   ▼
        ┌──────────────────────────────────────────────────────────┐
        │  STAGE 4  · Analysis & Figures                           │
        │  Code gen → execute → debug → validate                   │
        └──────────────────────────────────────────────────────────┘
                                   │
                                   ▼
        ┌
```

```
│  STAGE 5 · Paper Drafting                          │
└──────────────────────────────────────────────────┘
                          │
                          ▼
┌──────────────────────────────────────────────────┐
│  STAGE 6 · AI Peer Review Loop                    │
│  Reviewer → Revision Agent → iterate              │
└──────────────────────────────────────────────────┘
                          │
                          ▼
                [COMPLETED RUN LOG]
                          │
                          ▼
            ┌──────────────────────────────┐
            │  Continuous Learning Engine  │
            │  (see §10 for full spec)     │
            └──────────────────────────────┘
```

2.2 Agent Roster

Agent	Role	Model & Mode
Literature Scout	Searches, ranks, downloads academic papers. Reads PDFs and extracts structured findings.	Claude Code (computer use + file ops)
Feasibility Assessor	Synthesizes literature into go/no-go memo with confidence score.	Claude Opus, temp 0.2
Data Hunter	Identifies datasets, checks open access, generates download scripts, or triggers UI navigation.	Claude Code (headless + computer use)
Analysis Engine	Writes, runs, debugs, and validates analysis code. Generates figures to journal spec.	Claude Code (code execution)
Paper Writer	Drafts all manuscript sections. Inserts citations, formats to target style.	Claude Opus, temp 0.4
Reviewer Agent	Critiques paper against rubric. Returns structured revision requests.	Claude Opus, temp 0.05 (adversarial)
Revision Agent	Executes reviewer suggestions. Re-runs analysis if needed.	Claude Opus + Claude Code
Credential Broker	Manages the single human-in-the-loop credential touchpoint.	Python service

Learning Engine	Analyzes all completed runs. Updates prompts, strategies, and the Human Idea Model.	Claude Opus, scheduled nightly

3. Stage-by-Stage Specification

Stage 1 · Literature Scout

Database	Access Method
arXiv	Open API — no credentials required.
PubMed / PubMed Central	Open API (NCBI E-utilities). Full-text available for PMC papers.
Semantic Scholar	Open API (S2 v1). Returns abstracts, citation counts, PDF links.
Google Scholar	Headless browser (computer use). Rate-limited, used as fallback.
Unpaywall	Open API. Finds legal open-access PDF for any DOI.
Institutional proxy	Credential Broker supplies library login for Scopus, WoS, journal portals.

- Top 30 papers proceed to download; top 5 flagged as 'core references'
- Each paper extracted to structured JSON: claims, methods, findings, datasets used, relevance score

Stage 2 · Feasibility Assessment

- Go/no-go decision with confidence score 0–100. Below 40 → archive. Above 65 → full proceed.
- Output includes: novel angle, required datasets, recommended methods, estimated effort

Stage 3 · Data Acquisition

- Open path: OSF, Zenodo, Harvard Dataverse, UCI ML Repo, Kaggle public, Our World in Data
- Paywalled path: Credential Broker requests login once, stores encrypted, reuses forever
- Claude Code computer-use navigates portal UI: login → filter → download → verify integrity

Stage 4 · Analysis & Figures

- Analysis plan generated before any code is written
- Code generation loop with 5-retry auto-debug. Captures stdout/stderr/exit code each run.
- All figures at 300 DPI, colorblind-safe palette, generating code saved alongside output
- Full re-run from scratch as reproducibility check before proceeding to drafting

Stage 5 · Paper Drafting

- Section order: Methods → Results → Discussion → Introduction → Abstract → Title variants
- All references DOI-verified; BibTeX auto-generated in target journal style

Stage 6 · AI Peer Review Loop

- Reviewer Agent is explicitly adversarial — different system prompt from Writer, no shared memory
- 8-dimension rubric: novelty, statistical rigor, clarity, reproducibility, figure quality, abstract accuracy, overreach detection, reference quality
- Revision loop runs up to MAX_REVISIONS (default 4). Quality gate: score ≥ 82/100.
- After MAX_REVISIONS, best version flagged for human review with full revision history

4. Human Touchpoints

The pipeline is designed for zero-human-intervention operation. There are exactly five defined scenarios where human input may be requested.

Touchpoint	Description
1 · Credential Provision (BLOCKING)	Triggered once per institution when no stored credential exists. Human provides login via CLI. Stored AES-256 encrypted locally. Never triggered again for the same resource.
2 · CAPTCHA / Consent Form (BLOCKING)	Claude Code computer-use encounters an interactive wall it cannot automate. Human shown a screenshot and asked to complete it manually. Pipeline resumes automatically.
3 · Final Submission Approval (NON-BLOCKING)	Human receives final paper PDF + review history + quality score. 72-hour timeout before archiving as 'pending approval'.

4 · Quality Threshold Failure (NON-BLOCKING)	Paper fails to reach quality threshold after MAX_REVISIONS. Human can override, abandon, or trigger additional passes.
5 · Unresolvable Analysis Error (NON-BLOCKING)	Script fails 5 retries. Human notified. Pipeline proceeds with available results. Human can manually fix and trigger re-run.

> Design target: For a typical research idea with publicly available data, the pipeline runs end-to-end with ZERO human touchpoints. Credential provision is a one-time cost per institution.

5. Output Structure & Quality Standards

5.1 Final Deliverables per Run

Deliverable	Description
manuscript.pdf	Camera-ready paper formatted to target journal style.
manuscript.md	Editable Markdown source. Re-compilable to any format.
figures/ directory	All figures as PDF + PNG + generating code. 100% reproducible.
data/processed/	Cleaned dataset with data dictionary and provenance metadata.
references.bib	Complete BibTeX with DOI-verified entries.
execution_trace.jsonl	Structured log of every action taken by every agent.
review_history.jsonl	All reviewer reports and revision diffs across iterations.
quality_score.json	Final paper quality score + dimension breakdown.
learning_signal.json	NEW: structured signal consumed by Learning Engine after run.

5.2 Quality Gates

Gate	Criterion
Reviewer Score	Overall score ≥ 82/100
Figure Completeness	All planned figures generated with valid output files

Citation Coverage	≥ 80% of core references cited in paper
Statistical Validity	All key tests have p-values, effect sizes, confidence intervals
Abstract Match	Abstract claims supported by at least one result in paper
Word Count	Within ±15% of target journal word limit
Reproducibility	Full analysis re-runs from scratch with exit code 0

6. Configuration Reference

```
# Digital Unconscious — Research Pipeline Config

pipeline:
  mode: "claude_code"          # "claude_code" | "api"
  max_revisions: 4
  quality_threshold: 82
  auto_submit: false

literature:
  max_papers: 30
  core_papers: 5
  databases: [arxiv, pubmed, semantic_scholar, unpaywall]
  institutional_fallback: true

data:
  preferred_sources: [osf, zenodo, harvard_dataverse, uci_ml_repo]
  credential_vault: "~/.du/credentials.enc"

analysis:
  language: "python"
  figure_dpi: 300
  figure_format: [pdf, png]
  color_palette: "viridis"
  max_retries: 5
  timeout_seconds: 300

paper:
  target_journals:
    - "Nature Human Behaviour"
    - "PLOS ONE"
    - "arXiv preprint"
  citation_style: "apa"

agents:
  writer_temperature: 0.4
  reviewer_temperature: 0.05
```

```
  feasibility_temperature: 0.2

# ── Learning Engine config (see §10) ──────────────────────────────
learning:
  enabled: true
  schedule: "nightly"          # when to run the learning pass
  prompt_evolution: true       # allow prompts to be auto-updated
  human_idea_model: true       # build and update user idea profile
  min_runs_before_update: 3    # don't update prompts from < 3 runs
  evolution_temperature: 0.3   # temperature for meta-reasoning
  backup_prompts: true         # always keep previous prompt versions
```

7. Running the Pipeline

7.1 Automatic Trigger

```
du research --idea-id "uuid" --auto
```

7.2 Manual Commands

```
du research --idea "Cognitive load as SaaS pricing axis"
du research --idea-id abc123 --dry-run
du research --resume --run-id "run_2026_03_28_001"
du research --status
du research --logs --run-id run_001 --follow
```

8. Known Risks & Mitigations

Risk	Mitigation
Library portal UI changes break navigation	Navigation scripts are modular and logged. After 3 failures, human is notified with screenshot. Community can submit updated navigation flows.
Analysis generates statistically incorrect results	Reviewer Agent includes explicit statistical validity checks. Robustness checks run by default. All code and outputs logged.
Review loop never	Hard cap of MAX_REVISIONS. Best version flagged for human

converges	review with full history.
Dataset licensing prohibits redistribution	Data provenance captures license for every file. Data availability statement auto-generated. Proprietary data never committed to git.
Claude Code rate limits interrupt long run	Pipeline checkpoints after every stage. On rate-limit error, job queued and resumed. --session-id maintains context across interruptions.
Paper contains hallucinated citations	DOI verification runs on every reference. Unresolvable DOIs trigger re-search. Human sees unresolved count in approval notification.
Learning Engine degrades prompt quality	Every prompt evolution is versioned. Automatic rollback if new prompt produces lower average quality score on next 3 runs.

9. Development Roadmap

Milestone	Scope & Deliverables
M1 · Literature Scout (Weeks 1-2)	arXiv, PubMed, Semantic Scholar search + PDF download + structured extraction.
M2 · Feasibility Assessor (Week 3)	Synthesis prompt + go/no-go logic + integration with idea archive.
M3 · Data Acquisition Open (Weeks 4-5)	OSF, Zenodo, Dataverse connectors + data validation + provenance logging.
M4 · Credential Broker + UI (Weeks 6-7)	Encrypted vault + Claude Code computer-use for 3 major library portals.
M5 · Analysis Engine (Weeks 8-9)	Code generation loop + auto-debug + figure generation to journal spec.
M6 · Paper Writer (Weeks 10-11)	Section drafting + citation management + journal style formatting.
M7 · Review Loop (Week 12)	Adversarial Reviewer Agent + revision loop + quality scoring.
M8 · Learning Engine v1 (Weeks 13-14)	Run Outcome Analyzer + Human Idea Model v1 + Prompt Evolution. First learning cycle.
M9 · Integration & Release (Weeks 15-16)	Full integration with parent system. End-to-end test on 3 real ideas. GitHub release.

10. Continuous Learning Engine

> Core idea: every completed research run — whether it succeeds brilliantly or fails quietly — is a training signal. The Learning Engine studies these signals overnight and feeds improvements
> back into every part of the system: the agents' prompts, the RAG knowledge base, the idea scoring
> model, and a deepening model of how this specific human thinks.
>
> The system does not just get better at research in general.
> It gets better at research for you specifically.

10.1 Architecture Overview

```
Every completed pipeline run produces a learning_signal.json.
The Learning Engine runs nightly on the accumulated signals.

learning_signal.json  ──►  Run Outcome Analyzer
                                  │
                     ┌────────────┼──────────────┐
                     ▼            ▼              ▼
            Prompt Evolution   Domain Knowledge   Human Idea Model
            Engine             Expander           Builder
                     │            │              │
                     ▼            ▼              ▼
            Updated agent      Enriched RAG       Updated idea
            prompt library     knowledge base     fingerprint
                     │            │              │
                     └────────────┼──────────────┘
                                  │
                                  ▼
                         Meta-Learning Scheduler
                         (decides WHEN to apply updates,
                          runs A/B quality checks,
                          rolls back if quality drops)
```

10.2 Run Outcome Analyzer

After every completed run, the system records a structured learning signal capturing what

worked, what failed, and why. Over time, patterns emerge that allow the system to make smarter decisions early in future runs.

```
# learning_signal.json — produced after every run
{
  "run_id": "run_2026_03_28_001",
  "idea_id": "uuid",
  "idea_text": "Cognitive load as SaaS pricing axis",
  "idea_source_behaviors": ["browsed Notion pricing 8min", "searched 'SaaS
churn'"],
  "domain": "product design",

  "stage_outcomes": {
    "literature_scout": {
      "papers_found": 30,
      "papers_useful": 18,
      "best_database": "semantic_scholar",
      "worst_database": "google_scholar",
      "search_queries_that_worked": ["cognitive load measurement", "SaaS
pricing willingness to pay"],
      "search_queries_that_failed": ["mental effort pricing model"]
    },
    "feasibility": {
      "decision": "PROCEED",
      "confidence": 78,
      "was_decision_correct": true    # inferred from final paper quality
    },
    "data_acquisition": {
      "open_access_success": true,
      "credential_needed": false,
      "datasets_used": ["osf:cognitive_load_study_2023"],
      "datasets_that_failed": ["kaggle:saas_churn_private"]
    },
    "analysis": {
      "code_errors_encountered": 2,
      "auto_fixed": 2,
      "human_escalated": 0,
      "methods_that_worked": ["OLS regression", "bootstrap CI"],
      "methods_that_failed": []
    },
    "paper": {
      "sections_rewritten": ["Introduction", "Discussion"],
      "sections_accepted_first_draft": ["Methods", "Results"]
    },
    "review_loop": {
      "iterations": 3,
      "initial_score": 61,
      "final_score": 84,
      "most_common_critique_type": "statistical_rigor",
      "revisions_that_helped_most": ["add confidence intervals", "clarify
causal claims"]
    }
  },
```

```
  "final_quality_score": 84,
  "submitted": false,
  "human_feedback": null
}
```

The Run Outcome Analyzer aggregates these signals across all runs and identifies actionable patterns:

Pattern Type	Example Insight Extracted
Database effectiveness	Semantic Scholar consistently finds 40% more relevant papers than Google Scholar for this user's domains. → Reorder database priority.
Search query patterns	Queries using 'empirical study of X' outperform 'X research' by 2.3× relevance. → Update Scout's query generation prompt.
Analysis failure modes	pandas SettingWithCopyWarning causes 60% of auto-fix retries. → Pre-inject standard fix into all generated code.
Review bottlenecks	Statistical rigor critique appears in 80% of first reviews. → Add a statistical pre-check pass before the main review loop.
Feasibility calibration	Ideas with confidence 65-75 produce the best papers (high novelty, sufficient evidence). Recalibrate the scoring thresholds.
Section difficulty	Introduction rewrites occur 3× more often than other sections. → Use a stronger planning step before drafting the Introduction.

10.3 Human Idea Model

This is the most distinctive component of the Learning Engine. It builds a persistent, evolving model of how this specific human thinks — their intellectual fingerprint. Every behavior observation, every idea that gets generated, and every research outcome is a data point.

> Philosophical framing: A human's ideas are not random. They reflect deep patterns — recurring obsessions, preferred analogical domains, characteristic blind spots, and a distinctive style of connecting concepts. The Human Idea Model attempts to make these patterns explicit and use them to generate better-targeted ideas and research directions.

10.3.1 What the Model Tracks

```
# human_idea_model.json — updated nightly
{
  "user_id": "local_only_never_uploaded",
  "model_version": 47,           # increments every update cycle
  "last_updated": "2026-03-28",

  # ── Intellectual Fingerprint ──────────────────────────────────────
  "core_obsessions": [
    {"theme": "human cognition and AI interface", "strength": 0.91, "trend":
"stable"},
    {"theme": "pricing psychology", "strength": 0.74, "trend": "growing"},
    {"theme": "passive data collection systems", "strength": 0.68, "trend":
"new"}
  ],

  "preferred_analogy_domains": [
    "neuroscience → product design",
    "economics → UX",
    "biology → system architecture"
  ],

  "cognitive_style": {
    "divergence_preference": 0.82,      # prefers many ideas over deep single
ideas
    "abstraction_level": "mid-high",    # prefers frameworks over pure theory
or pure practice
    "evidence_threshold": "moderate",   # does not require exhaustive proof
before acting on idea
    "novelty_vs_rigor_tradeoff": 0.65   # leans toward novelty (1.0 = pure
novelty, 0.0 = pure rigor)
  },

  # ── Behavioral Timing Patterns ────────────────────────────────────
  "peak_creative_windows": [
    {"time": "09:00-11:00", "day": "weekday", "idea_quality_avg": 0.84},
    {"time": "21:00-23:00", "day": "any",     "idea_quality_avg": 0.79}
  ],
  "low_signal_windows": ["13:00-14:30"],  # post-lunch, ideas rarely useful

  # ── Domain Attention Map ──────────────────────────────────────────
  "domain_attention": {
    "product_design":    {"screen_time_fraction": 0.31,
"idea_conversion_rate": 0.44},
    "ai_tools":          {"screen_time_fraction": 0.28,
"idea_conversion_rate": 0.51},
    "cognitive_science": {"screen_time_fraction": 0.12,
"idea_conversion_rate": 0.67},
    "business_models":   {"screen_time_fraction": 0.18,
"idea_conversion_rate": 0.29}
  },

  # ── Cross-Domain Connection Patterns ──────────────────────────────
```

```
  "productive_crossings": [
    {"from": "cognitive_science", "to": "product_design", "yield": 0.82},
    {"from": "ai_tools",          "to": "business_models", "yield": 0.71},
    {"from": "biology",           "to": "system_design",   "yield": 0.63}
  ],
  "unproductive_crossings": [
    {"from": "finance", "to": "design", "yield": 0.12}
  ],

  # ── Idea Lifecycle Patterns ──────────────────────────────────────
  "ideas_generated_total": 847,
  "ideas_reaching_research": 23,
  "ideas_completing_paper": 11,
  "idea_to_paper_conversion": 0.013,

  "what_makes_a_good_idea_for_this_user": [
    "Combines an observable human behavior pattern with a measurable outcome",
    "Has a clear 'why hasn't this been studied yet' angle",
    "Can be tested with publicly available behavioral or usage data",
    "Sits at the boundary of 2+ of user's core obsession domains"
  ],

  # ── Blind Spots Identified ───────────────────────────────────────
  "recurring_blind_spots": [
    "Underestimates data availability for behavioral economics studies",
    "Overestimates novelty of ideas in HCI domain — often rediscovery",
    "Rarely generates ideas involving longitudinal study designs"
  ],

  # ── Evolving Research Taste ──────────────────────────────────────
  "research_taste_trajectory": [
    {"period": "2025-Q3", "dominant_theme": "AI interfaces"},
    {"period": "2025-Q4", "dominant_theme": "behavioral economics"},
    {"period": "2026-Q1", "dominant_theme": "passive intelligence systems"}
  ]
}
```

10.3.2 How the Model Improves the System

Downstream Effect	How the Human Idea Model Feeds Into It
Idea Generation Prompt	The Creative Agent receives the user's core obsessions, preferred analogy domains, and productive crossing patterns as context. Ideas it generates are pre-filtered to match the user's intellectual style.
Judge Agent Scoring	The Judge recalibrates its scoring rubric based on what kinds of ideas from this user have historically led to good papers. The rubric is personalized, not generic.

Literature Scout Queries	Search query templates are enriched with the user's productive domain crossings, surfacing papers the generic query would miss.
Peak Window Scheduling	The system schedules its most expensive idea generation passes during the user's historically high-quality creative windows.
Blind Spot Injection	Before each research run, a brief prompt reminds the agents of the user's known blind spots, counterbalancing them explicitly.
RAG Priority	Domains with high idea_conversion_rate get proportionally larger knowledge base allocations during RAG construction.

10.4 Prompt Evolution Engine

Every agent in the pipeline uses a versioned prompt. The Prompt Evolution Engine analyzes performance data across runs and proposes incremental improvements to each prompt. It is the system teaching itself to be a better researcher.

> Design constraint: Prompt Evolution is conservative by design.
> Prompts are updated incrementally, never rewritten wholesale.
> Every update is versioned. Automatic rollback triggers if quality drops.
> The system never evolves a prompt based on fewer than 3 completed runs.

```
# Prompt Evolution cycle — runs nightly after Run Outcome Analyzer

for each agent_prompt in [scout_prompt, feasibility_prompt,
                          writer_prompt, reviewer_prompt, ...]:

    # 1. Gather evidence from recent runs
    recent_failures = get_failure_patterns(agent, last_n_runs=10)
    recent_successes = get_success_patterns(agent, last_n_runs=10)

    # 2. Ask Claude Opus to diagnose the prompt
    diagnosis = claude_opus(
        system="You are a prompt engineer analyzing agent performance data.",
        user=f"""
        Current prompt for {agent.name}:
        {agent.current_prompt}

        Recent failure patterns:
        {recent_failures}

        Recent success patterns:
        {recent_successes}
```

```
        Identify the single most impactful change to this prompt
        that would reduce the most common failure pattern.
        Return a minimal, surgical edit — not a rewrite.
        Explain your reasoning.
        """,
        temperature=0.3
    )

    # 3. Propose and version the change
    proposed_prompt = apply_edit(agent.current_prompt, diagnosis.edit)
    version = increment_version(agent.prompt_version)
    save_prompt(agent, proposed_prompt, version)

    # 4. Shadow-run new prompt on 3 historical inputs
    shadow_scores = [evaluate(proposed_prompt, historical_input)
                     for historical_input in get_test_cases(agent, n=3)]

    # 5. Apply only if shadow run shows improvement
    if mean(shadow_scores) > agent.baseline_score:
        activate_new_prompt(agent, version)
        log_evolution(agent, version, diagnosis.reasoning)
    else:
        discard(proposed_prompt, version)
        log_rejection(agent, version, diagnosis.reasoning, shadow_scores)
```

10.4.1 Prompt Version History

```
# ~/.du/prompts/
  reviewer_agent/
    v1.0.0_original.txt
    v1.1.0_added_ci_check.txt       # evolution: added explicit CI instruction
    v1.2.0_overreach_detector.txt   # evolution: added causal claim detection
    v1.2.1_rollback.txt             # ROLLBACK: v1.2.0 increased false
positives
    v1.3.0_current.txt
    evolution_log.jsonl              # every change, reason, and quality delta
```

10.5 Domain Knowledge Expander

The RAG knowledge base is not static. The Domain Knowledge Expander automatically enriches it based on what the system learns is useful for this user's research patterns.

- After each run, highly-cited papers from successful literature scouts are added to the permanent RAG base
- Papers that were retrieved but never cited in any final paper are downweighted

- The expander identifies adjacent domains that appeared in successful crossings and fetches foundational papers from those domains proactively
- Domain knowledge decay: papers older than 5 years in fast-moving fields (AI, ML) are flagged for freshness checking and replacement

```
# Domain Knowledge Expander — nightly run
{
  "additions_this_cycle": [
    {"paper": "doi:10.xxxx/cognitive_load_2024", "reason": "cited in 3 of last
5 papers"},
    {"paper": "doi:10.xxxx/saas_churn_2025",     "reason": "appeared in
productive crossing: pricing×psychology"}
  ],
  "downweighted_this_cycle": [
    {"paper": "doi:10.xxxx/old_ui_study_2018",   "reason": "retrieved 6 times,
cited 0 times"}
  ],
  "proactive_additions": [
    {"domain": "behavioral economics",  "papers_added": 12,
     "reason": "high idea_conversion_rate (0.67), knowledge base
underpopulated"}
  ],
  "freshness_flags": [
    {"paper": "doi:10.xxxx/llm_survey_2022", "reason": "AI domain, >2 years
old, newer surveys exist"}
  ]
}
```

10.6 Meta-Learning Scheduler

The Meta-Learning Scheduler is the coordinator that decides when and how aggressively to apply learning updates. Its primary job is to prevent the system from over-fitting to a small number of runs and degrading.

Scheduling Rule	Rationale
Never update a prompt from fewer than 3 completed runs	Protects against one-run anomalies corrupting a good prompt
Apply at most one prompt evolution per agent per week	Prevents oscillation — changes need time to demonstrate their effect
Run shadow evaluation on 3 historical inputs before activating	Catches regressions before they affect real runs

Rollback if next 3 runs after update show quality drop > 5 points	Automatic reversion without human intervention
Freeze all evolution if overall quality score drops 2 weeks in a row	Circuit breaker — escalates to human for diagnosis
Log all decisions with full reasoning in evolution_log.jsonl	Auditability — human can always review what the system did to itself

10.7 Learning Engine Output — What Changes Over Time

After 30 days of use, a typical user should observe the following improvements compared to day 1:

Metric	Expected Improvement After 30 Days
Idea-to-paper conversion rate	+40–60%. The system generates fewer ideas but they are better matched to what this user can actually execute into good research.
Literature scout precision	+30%. Search queries have been refined toward this user's productive domains. Fewer irrelevant papers downloaded.
First-draft review score	+8–12 points. Writer Agent's prompt has been evolved to pre-empt the most common critique types for this user's style.
Review loop iterations	−1 iteration on average. Statistical rigor and clarity issues are caught earlier.
Analysis auto-fix success rate	+20%. Common code error patterns specific to this user's preferred data types are pre-empted in generated code.
RAG relevance	+35%. Knowledge base is now dense in the user's productive crossing domains, sparse in unproductive ones.

> Compounding effect: because every component learns simultaneously and feeds back into the others,
> improvement is not linear — it accelerates. A system that has processed 100 research runs is not just 100× better than day 1. It has developed a coherent model of a specific human mind

> that no general-purpose AI system can replicate from scratch.
>
> This is the moat. Not the technology — the accumulated understanding of you.

10.8 Privacy & Ownership of the Learning Data

Principle	Implementation
All learning data is local	human_idea_model.json, all prompt versions, evolution logs — stored exclusively on the user's device. Never sent to any server.
User owns the model	The Human Idea Model is a plain JSON file. Users can read it, edit it, delete it, export it, or fork it freely.
Opt-out granularity	Users can disable: prompt evolution only, idea modeling only, or the entire Learning Engine — without affecting the rest of the pipeline.
Transparency log	A plain-language summary of every change the Learning Engine made is written to learning_changes.md after each cycle. No black boxes.
No cross-user learning	The system learns only from this user's data. There is no federated learning, no model sharing, no phone-home.

```
# learning_changes.md — human-readable summary written after each cycle
# example entry:

## 2026-03-28  Learning Cycle #47

### Prompt Updates
- reviewer_agent: Added explicit instruction to check for missing confidence
  intervals on all regression coefficients. Triggered by: same critique
appeared
  in 4 of last 5 reviews. Shadow test: quality +3.2 points. Activated.

### Human Idea Model Updates
- New productive crossing identified: cognitive_science → pricing_psychology
  (2 ideas from this crossing both reached papers in the last 30 days)
- Peak creative window refined: 21:00–23:00 window confirmed high-quality
  (avg score 0.81 across 12 observations)
- New blind spot detected: ideas involving longitudinal designs are
consistently
  underestimated for data availability. Added reminder to feasibility prompt.

### Knowledge Base Updates
```

```
- Added 8 papers on behavioral pricing (high idea_conversion_rate domain)
- Downweighted 3 papers on generic UX (retrieved 4 times, cited 0 times)

### No changes to: data acquisition, analysis engine, paper writer
(insufficient data)
```

11. Long-Term Vision: The Working Persona

> After months of continuous operation, the system stops being a tool and becomes something else:
> an AI that knows how you think better than any general-purpose assistant ever could.
>
> It knows which of your ideas are half-baked enthusiasm versus genuine insight.
> It knows which domains energize you versus which you browse out of obligation.
> It knows when you are most creative and when your attention is scattered.
> It knows which analytical methods you gravitate toward and which you avoid.
>
> This accumulated knowledge — the Human Idea Model — is your working persona:
> a digital twin of your intellectual patterns that amplifies your strengths,
> compensates for your blind spots, and generates research in your voice.

11.1 Flywheel Dynamics

```
More screen behavior observed
        │
        ▼
Richer Human Idea Model
        │
        ▼
Better idea generation  ──▶  More ideas reaching research
        │
        ▼
More completed research runs
        │
        ▼
More learning signals for Prompt Evolution Engine
        │
        ▼
Higher quality papers  ──▶  Higher Judge Agent scores
```

```
        |
        ▼
More ideas cleared for research  ──▶  back to top

The flywheel accelerates over time.
The system's value to you grows with every hour it runs.
```

11.2 What This Means for Creative Work

The conventional fear about AI in creative and research contexts is substitution: the AI replaces the human's role. This system is designed around a different model.

Conventional AI Tool	Digital Unconscious After 6 Months
Same output quality for every user	Output quality is specific to you — built from your unique behavior patterns
You must explain your context each session	The system already knows your context — it has been observing it for months
Generates generic ideas in your domain	Generates ideas at the intersection of your specific obsessions and blind spots
You judge quality	The system has learned your quality standards from 100+ examples of your past judgments
Replaces your thinking	Amplifies your thinking by externalizing and extending your unconscious patterns

> The output of this system is not research papers.
> The output is a progressively clearer mirror of how you think —
> and a machine that thinks alongside you in your own voice.